

# Search sheets: how to run the activity



The pages after this one are 24 **search sheets**, built from *Green Eggs and Ham* by Dr Seuss. Print them one-sided and hand out **one sheet per person** — every sheet is different, and nothing needs cutting out.

Each sheet holds 41–42 **token pairs**. A **token** is one word or one punctuation mark; a token pair is a **next token** paired with the **previous token** that came immediately before it somewhere in the original text.

In **not eat**, the boxed token is the previous token and *eat* is the next token. Each token keeps the same colour wherever it appears, so scan by colour first and read second. With only 12 colours and far more tokens than that, a colour narrows the search rather than settling it — always check the token itself.

993 tokens, 56 unique · 1.8 bits/token · perplexity 3.5 · branching factor 3.45

## Running a round

1. **Call out the last token** of the text so far.
2. Everyone scans their own sheet for token pairs whose previous token matches. **Hands up** — one hand each, and anyone with more than one match just goes with the first they spot.
3. **Pick a raised hand at random** and ask for that token pair's next token.
4. A scribe writes the token on the board. That token becomes part of the context for the next round. Repeat.

Picking at random matters. If you take whoever shouts first you are sampling the fastest reader, not the text — and the whole point is that the room samples the way the model does.

## Worked example

**Start anywhere.** Pick any token pair — say this one — and write its previous token *and* its next token on the board.

**Round 1.** Call out the last token on the board (*eat*). Another sheet has a match, so that person raises a hand and reads out *green*.

**Round 2.** Call out the last token on the board (*green*). Another sheet has a match, so that person raises a hand and reads out *eggs*.

Keep going for as long as you like. The text the room writes will sound like *Green Eggs and Ham* without ever repeating it — that is the model generating, not the corpus being read out.

## What to watch for

**The hands are the probability distribution.** Every token pair was dealt to exactly one person, and the deal spreads each context across as many people as it will go, so a common continuation really does put more hands in the air than a rare one. Nobody has to explain weighted sampling; the room performs it. (Roughly: for a context more common than the class is big, somebody holds two matches but raises one hand, which flattens the busiest contexts.)

**Nobody holds the model.** No single sheet can continue the text on its own. The model only exists across the whole room — and if somebody is away, their share of it is missing, and some contexts will draw no hands at all.

**Running dry.** If no hands go up, you have reached a context that only ever occurred at the very end of the text. Start again from any token pair.

*the token pair that answers*

*on the board*

**not** eat

not eat

**eat** green

not eat green

**green** eggs

not eat green eggs

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "not", then the token pair **not** eat is a match and you answer **eat**.

---

**not** eat

! would

car !

**Sam** I

green eggs

am Sam

and ham

I do

not ,

that Sam

, could

fox .

with a

them anywhere

. they

I am

I do

them .

like green

eggs and

will not

you eat

a train

rain .

do not

a box

do not

them in

? would

not like

here or

like them

in a

in a

mouse .

. I

eat them

you ,

may ,

. .

would not

. so

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "green", then the token pair **green** eggs is a match and you answer eggs.

---

am !

green eggs

not with

! Thank

eggs and

? I

not like

Sam !

. I

eat green

. .

a mouse

in the

them in

with a

car !

I do

, with

. I

you like

rain .

like green

them in

may like

I will

not like

in a

them ,

you may

do not

here or

fox .

do not

a tree

I like

mouse .

like them

would not

and ham

that Sam

will not

. say

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "not", then the token pair **not** on is a match and you answer **on**.

---

**not** on

. not

**may** like

**in** a

**car** .

**green** eggs

**them** with

**do** not

**would** not

**here** or

**with** a

**mouse** .

**not** in

**a** car

**will** see

**do** not

**see** .

**Sam** I

**in** the

**like** them

**I** do

. and

**you** say

**I** do

. I

**and** ham

**eggs** and

? I

**you** see

**fox** .

, **Sam**

**a** car

**eat** them

**that** Sam

**not** in

**them** in

. and

**them** here

**I** am

! you

**like** them

**rain** .

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "I", then the token pair **I** am is a match and you answer **am**.

---

**I** am

, you

**them** with

**do** so

**I** do

**a** train

**a** car

**see** .

**here** or

**not** like

**in** the

**you** may

**car** ?

**them** with

**like** green

**box** ?

**do** not

**with** a

**I** am

**on** a

**like** them

**fox** .

**eggs** and

**not** ,

**you** ,

**green** eggs

**eat** them

**Sam** I

**!** let

**.** I

**in** a

**and** I

**mouse** .

, so

**not** in

**they** are

**?** eat

**.** I

**would** you

**tree** !

**them** !

**.** I

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "I", then the token pair **I** would is a match and you answer **would**.

---

. I

I would

would you

, in

like them

Sam !

them ,

a train

you do

? not

or there

with a

, Sam

you ,

them ,

not on

box .

see .

here or

I do

car .

they are

. try

eat them

. I

fox ?

and in

like them

not like

do not

in a

tree .

green eggs

! I

I do

them with

on a

eggs and

do not

a train

in the

not eat

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "or", then the token pair **or there** is a match and you answer **there**.

---

**or** there

**in** a

**eggs** and

**so** like

**Sam** I

**.** try

**I** am

**them** anywhere

**box** .

**in** a

**here** and

**a** boat

**would** not

**and** you

**like** them

**eat** them

**not** like

**.** I

**them** here

**would** you

**!** I

**I** do

**tree** .

**see** !

**not** like

**you** do

**like** them

**I** am

**a** train

**on** a

**green** eggs

**,** could

**,** would

**them** in

**do** not

**not** like

**?** not

**car** !

**.** I

**are** .

**do** not

**you** ,

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "a", then the token pair **a house** is a match and you answer **house**.

---

**a** house

**are** so

**car** .

**let** me

**here** or

**and** on

**them** ,

**like** them

**on** a

**in** the

**you** see

**do** not

**would** eat

**,** could

**them** anywhere

**I** am

**not** in

**green** eggs

**like** them

**?** I

**!** I

**eat** them

**Sam** I

**not** like

**.** I

**do** not

**would** not

**a** fox

**,** could

**I** will

**dark** ?

**or** there

**them** in

**.** you

**in** a

**you** may

**tree** .

**.** I

**box** .

**not** like

**I** will

**so** good



When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "not", then the token pair **not** like is a match and you answer **like**.

---

**not** like

a train

**them** here

**could** you

**so** you

I do

**them** here

**eat** them

I will

! If

a boat

**do** not

**like** them

? could

**let** me

. I

I do

anywhere .

tree !

**in** the

**not** in

**would** you

**and** ham

**like** them

**you** ?

**dark** .

. say

**would** not

**box** .

. and

**green** eggs

**in** the

**or** there

**not** eat

. I

**here** in

**do** not

**he** tries

, Sam

**them** anywhere

**on** a

, with

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "in", then the token pair **in** a is a match and you answer **a**.

---

. and

in a

will not

like them

a car

do not

, Sam

. a

them !

them in

eat them

box .

let me

I do

, Sam

I am

and ham

do not

tree !

a car

you let

anywhere .

? I

dark !

could you

not in

the dark

! so

. and

would not

so good

green eggs

like them

not in

not like

. and

in the

I will

on a

or there

them ,

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "like", then the token pair **like** **green** is a match and you answer **green**.

---

**you** ,

**like** green

**green** eggs

**them** here

**eat** them

**not** ,

**a** house

**would** you

**!** Thank

**a** mouse

**so** I

**a** house

**I** would

**them** with

**the** dark

**,** in

**ham** !

**there** ?

**dark** .

**.** I

**.** not

**I** do

**,** Sam

**box** .

**.** not

**I** am

**or** there

**anywhere** .

**not** ,

**do** not

**.** and

**them** anywhere

**let** me

**?** I

**in** a

**in** a

**and** ham

**will** not

**not** like

**I** will

**could** not

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "like", then the token pair **like** **them** is a match and you answer **them**.

---

**dark** ?

. I

**them** ,

**like** **them**

**not** ,

. I

**a** **mouse**

! **you**

**and** I

**will** not

. I

**not** **like**

I **do**

**them** in

**not** ,

**you** **let**

? I

**them** !

, **could**

**could** **you**

**there** .

**in** a

**not** **like**

, **could**

**with** a

**a** **mouse**

I **will**

**house** .

**a** **box**

**anywhere** .

**am** .

**green** **eggs**

**would** **you**

**do** not

**ham** ?

**eat** **green**

I **am**

. not

**or** there

I **will**

**the** **rain**

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "me", then the token pair **me** be is a match and you answer **be**.

---

**me** be

**not** like

**I** do

, **could**

. **not**

! **not**

**them** in

**not** ,

. .

. I

**I** would

**I** do

. **would**

a **tree**

a **box**

a **mouse**

**like** that

**not** with

**could** not

in a

, **Sam**

**do** not

**house** .

**will** not

**I** do

**would** you

**there** .

**am** !

**eat** them

**and** he

**with** a

**the** rain

**dark** .

**or** there

**them** here

**not** like

**ham** !

**anywhere** .

**you** will

! **could**

**them** .

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "like", then the token pair **like** **them** is a match and you answer **them**.

---

**like** **them**

, **on**

. **I**

! **that**

**a** **mouse**

! **I**

**would** **you**

**will** **eat**

**eat** **them**

**could** **not**

**a** **goat**

**not** **like**

**I** **do**

**house** ?

**not** **eat**

**ham** .

**I** **would**

**Sam** **I**

**not** **in**

**a** **fox**

, **would**

**dark** !

**them** **here**

**in** **a**

**and** **ham**

. **I**

**them** **and**

**me** **be**

**there** .

**I** **do**

. **not**

**I** **do**

**not** **like**

**the** **dark**

**am** .

. **I**

**them** **in**

**with** **a**

**you** **like**

**do** **not**

**anywhere** .

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "and", then the token pair **and** **I** is a match and you answer **I**.

---

**,** **you**                      **and** **I**                      **a** **box**                      **ham** .

**I** **would**                      **like** **them**                      **not** **like**                      **house** .

**I** **would**                      **.** **I**                      **could** **not**                      **I** **will**

**anywhere** !                      **I** **do**                      **.** **would**                      **Sam** ,

**will** **not**                      **,** **would**                      **not** ,                      **.** **would**

**am** .                      **them** !                      **!** **I**                      **with** **a**

**like** **green**                      **eat** **them**                      **.** **I**                      **not** **like**

**a** **house**                      **them** **anywhere**                      **would** **you**                      **not** ,

**you** **on**                      **and** **in**                      **me** **be**                      **in** **a**

**do** **not**                      **!** **I**                      **there** .                      **a** **goat**

**the** **dark**

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "a", then the token pair **a house** is a match and you answer **house**.

---

**!** not

**a** house

**.** I

**,** could

**ham** .

**will** not

**do** not

**house** .

**would** not

**like** them

**there** .

**.** not

**I** am

**a** house

**could** not

**them** anywhere

**I** would

**with** a

**not** like

**not** eat

**not** eat

**in** the

**the** dark

**and** there

**like** them

**me** be

**you** ,

**!** try

**not** like

**.** I

**a** goat

**.** you

**and** ham

**I** would

**eat** them

**,** in

**Sam** I

**them** in

**am** .

**I** am

**anywhere** !



When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "the", then the token pair **the** dark is a match and you answer **dark**.

---

**the** dark

**house** .

**goat** !

**.** I

**.** and

**do** not

**.** you

**could** you

**them** in

**train** .

**in** a

**,** on

**say** .

**and** in

**am** .

**!** Sam

**like** them

**!** in

**there** .

**a** car

**Sam** I

**with** a

**will** eat

**I** do

**not** like

**like** them

**would** you

**you** ,

**.** I

**a** tree

**I** will

**not** in

**a** fox

**them** with

**eat** them

**not** ,

**,** could

**ham** ?

**not** in

**I** would

**I** say

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "like", then the token pair **like** green is a match and you answer green.

---

**like** green

**in** the

**I** do

**.** not

**with** a

**.** that

**like** them

**not** like

**could** not

**am** .

**not** like

**!** eat

**and** you

**ham** .

**there** .

**a** box

**would** you

**not** eat

**Sam** .

**a** goat

**,** in

**not** like

**will** eat

**a** mouse

**them** with

**train** !

**house** .

**them** with

**do** not

**I** do

**eat** them

**!** a

**I** will

**I** do

**.** and

**,** will

**say** !

**the** rain

**.** could

**goat** ?

**you** ,

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "like", then the token pair **like** **them** is a match and you answer **them**.

---

**like** **them**

**a** **tree**

**would** **not**

**not** **eat**

**not** **in**

**with** **a**

**them** **,**

**them** **,**

**I** **do**

**.** **I**

**do** **not**

**.** **not**

**not** **like**

**in** **a**

**house** **.**

**.** **you**

**could** **you**

**train** **.**

**will** **eat**

**,** **with**

**ham** **!**

**not** **eat**

**I** **could**

**the** **rain**

**I** **would**

**!** **and**

**and** **I**

**!** **would**

**I** **will**

**like** **them**

**there** **.**

**goat** **.**

**Sam** **.**

**a** **fox**

**a** **train**

**.** **would**

**you** **like**

**say** **.**

**,** **Sam**

**eat** **them**

**am** **Sam**

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "I", then the token pair **I will** is a match and you answer **will**.

---

**I** will

. I

**not** like

**ham** .

**eat** them

. not

**say** !

**train** !

**be** .

**the** dark

**I** would

**like** them

. I

**will** not

**you** may

, I

**I** am

**do** not

**them** .

**like** them

**am** .

. .

**with** a

**Sam** I

**them** in

**not** in

**could** you

**and** ham

**I** do

**a** mouse

**a** house

**a** car

**not** like

**not** ,

, **Sam**

**would** not

! and

**in** the

**goat** .

! I

**in** a

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "do", then the token pair **do** not is a match and you answer **not**.

---

**do** not

**them** with

**I** am

**.** I

**will** eat

**tries** them

**I** will

**!** would

**not** with

**you** ,

**and** ham

**?** in

**like** them

**a** box

**in** a

**not** in

**ham** .

**I** do

**not** like

**.** I

**.** I

**,** in

**boat** .

**!** here

**would** not

**be** !

**like** green

**.** and

**not** like

**a** boat

**Sam** I

**I** like

**eat** them

**in** a

**am** !

**eggs** and

**say** !

**train** !

**with** a

**them** with

**could** not

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "eggs", then the token pair **eggs** **and** is a match and you answer **and**.

---

**eggs** **and**

**train** **!**

**If** **you**

**be** **!**

**not** **eat**

**.** **I**

**and** **I**

**Sam** **I**

**will** **eat**

**them** **anywhere**

**a** **box**

**,** **Sam**

**them** **here**

**do** **not**

**?** **would**

**I** **will**

**.** **I**

**I** **do**

**could** **not**

**.** **.**

**eat** **them**

**not** **with**

**in** **a**

**not** **,**

**mouse** **.**

**!** **not**

**a** **fox**

**would** **you**

**.** **I**

**I** **do**

**try** **them**

**like** **them**

**am** **.**

**!** **a**

**I** **am**

**in** **a**

**you** **?**

**not** **eat**

**with** **a**

**boat** **.**

**like** **green**

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "like", then the token pair **like** **them** is a match and you answer **them**.

---

**like** them

**you** will

**a** train

**Thank** you

**boat** ?

**in** a

**,** could

**.** say

**.** Sam

**not** ,

**.** not

**will** try

**and** ham

**them** with

**I** would

**Sam** I

**not** with

**eggs** and

**do** not

**in** a

**a** fox

**be** ,

**.** I

**?** I

**try** them

**do** not

**!** I

**eat** them

**train** ?

**not** ,

**you** like

**!** a

**here** they

**with** a

**them** here

**am** !

**I** would

**mouse** .

**I** am

**like** them

**would** eat

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "you", then the token pair **you** eat is a match and you answer **eat**.

---

**you** eat

**I** do

**,** Sam

**with** a

**eat** them

**.** would

**.** and

**not** like

**try** them

**them** in

**not** in

**Sam** I

**Thank** you

**!** I

**do** !

**mouse** .

**you** let

**will** eat

**and** I

**a** house

**a** tree

**?** here

**eggs** and

**like** them

**good** ,

**like** them

**train** !

**I** will

**in** a

**them** .

**in** a

**would** not

**do** not

**fox** .

**not** in

**I** do

**.** not

**here** or

**!** not

**.** I

**am** .



When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "would", then the token pair **would** not is a match and you answer **not**.

---

**would** not

**like** them

. Sam

**fox** .

**in** a

**may** .

. not

**I** do

**I** would

**in** a

, I

**do** not

**mouse** ?

**not** like

**rain** ?

**not** like

**not** ,

**eat** them

**them** ,

**try** them

**here** or

**eggs** and

**do** not

**Sam** I

**like** green

**will** see

. you

! you

? I

. I

**good** ,

**a** fox

**a** tree

**am** .

! I

**and** I

**you** ,

**I** do

**with** a

**you** !

**them** ,